

Flutter

1. Define Flutter and write the steps to install Flutter SDK in Ubuntu.

Flutter is an open-source UI software development kit (SDK) created by Google. It is used for building natively compiled, multi-platform applications from a single codebase. This means developers can write one set of code and deploy it to various platforms, including mobile (Android and iOS), web, desktop (Windows, macOS, and Linux), and even embedded devices.

Steps to install Flutter SDK:

1. Installing Flutter on Linux

Open your terminal and execute the following command to install Flutter:

```
$ sudo snap install flutter --classic
```

2. Verify Flutter Dependencies

Once the installation is finished, run the following command to perform various checks and assess the status of your Flutter installation and its dependencies, ensuring that your development environment is set up correctly.

```
$ flutter doctor
```

3. Installing Google Chrome on Linux

Let's install google chrome with this command.

```
sudo snap install chromium
```

chromium is a open-source browser, where any other browsers are build.

4. Installing and Configuring Android Studio on Linux

Android Studio can be installed in different ways that have been covered in our separate article on how to install Android Studio on Linux.

For a quick one-step process, just execute the following command in your terminal to effortlessly install Android Studio using Snap.

```
$ sudo snap install android-studio --classic
```

Once the installation is complete, launch Android Studio either by executing the following command or searching for it within your operating system's menu.

```
$ android-studio
```

2. Write a "Hello World" program using Stateless and Stateful widgets.

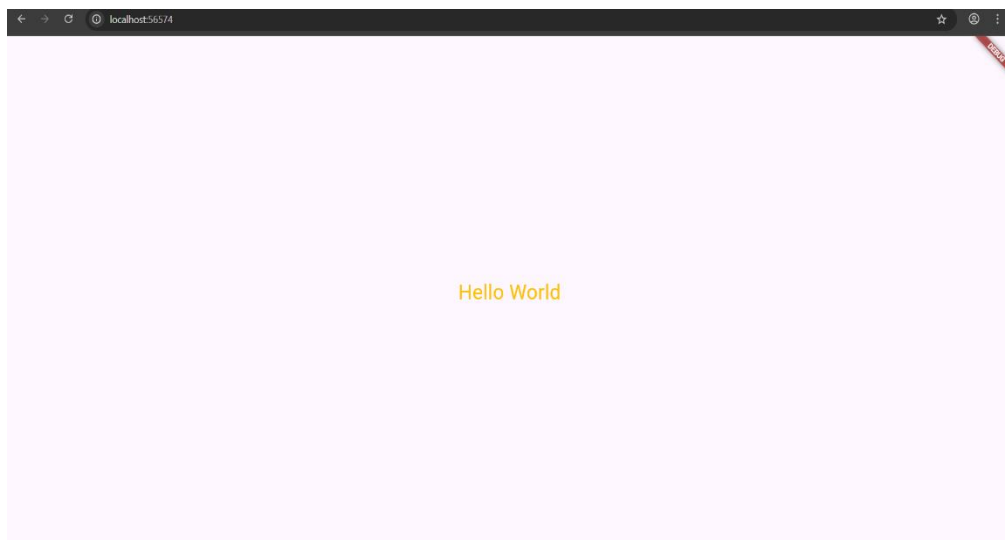
Code:

```
//Stateless Widget
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        body: Center(
          child: Text(
            'Hello World',
            style: TextStyle(fontSize: 30,color: Colors.amber),
          ),
        ),
      ),
    );
  }
}
```

Output:



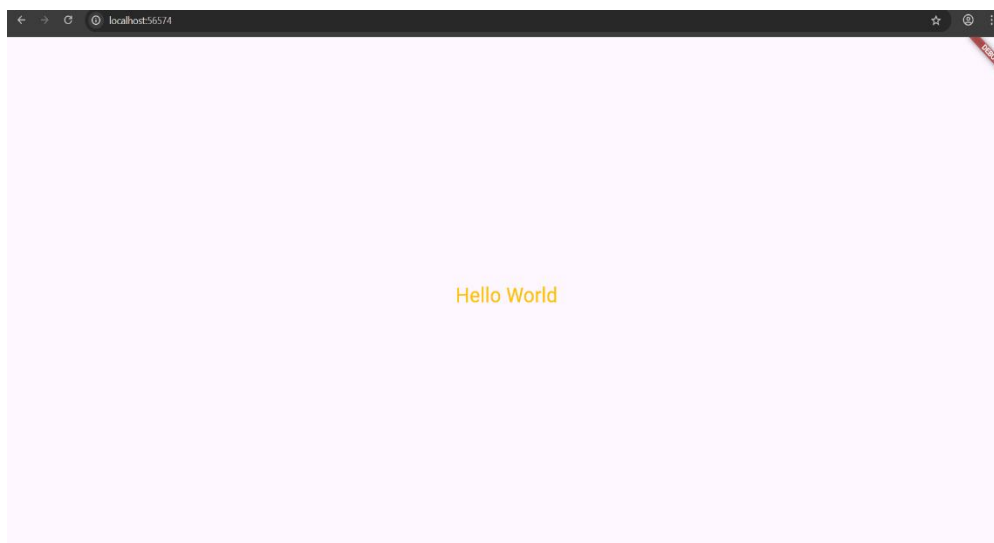
```
//Stateful Widget
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatefulWidget {
  @override
  State<MyApp> createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        body: Center(
          child: Text(
            'Hello World',
            style: TextStyle(fontSize: 30,color: Colors.amber),
          ),
        ),
      ),
    );
  }
}
```

Output:



3. Write an example program to demonstrate decision-making statements in Dart.

Code:

```
import 'dart:io';

void main() {
  print("Enter a number:");
  int num = int.parse(stdin.readLineSync()!);

  // if-else statement
  if (num > 0) {
    print("The number is positive.");
  } else if (num < 0) {
    print("The number is negative.");
  } else {
    print("The number is zero.");
  }

  // nested if statement
  if (num != 0) {
    if (num % 2 == 0) {
      print("It is an even number.");
    } else {
      print("It is an odd number.");
    }
  }

  // switch statement
  print("\nEnter a grade (A, B, C, D):");
  String grade = stdin.readLineSync()!.toUpperCase();
  switch (grade) {
    case 'A':
      print("Excellent!");
      break;
    case 'B':
      print("Good job!");
      break;
    case 'C':
      print("You passed.");
      break;
    case 'D':
      print("Try harder next time.");
      break;
    default:
      print("Invalid grade.");
  }
}
```

Output:

Enter a number:

5

The number is positive.

It is an odd number.

Enter a grade (A, B, C, D):

A

Excellent!

4. Write an example program to demonstrate looping statements in Dart.

Code:

```
import 'dart:io';

void main() {
  // for loop
  print("For Loop:");
  for (int i = 1; i <= 5; i++) {
    print("Number: $i");
  }

  // while loop
  print("\nWhile Loop:");
  int j = 1;
  while (j <= 5) {
    print("Count: $j");
    j++;
  }

  // do-while loop
  print("\nDo-While Loop:");
  int k = 1;
  do {
    print("Value: $k");
    k++;
  } while (k <= 5);

  // break and continue demonstration
  print("\nBreak and Continue Example:");
  for (int n = 1; n <= 10; n++) {
    if (n == 6) break; // exits loop when n = 6
    if (n % 2 == 0) continue; // skips even numbers
    print("Odd number: $n");
  }
}
```

Output:

For Loop:

Number: 1

Number: 2

Number: 3

Number: 4

Number: 5

While Loop:

Count: 1

Count: 2

Count: 3

Count: 4

Count: 5

Do-While Loop:

Value: 1

Value: 2

Value: 3

Value: 4

Value: 5

Break and Continue Example:

Odd number: 1

Odd number: 3

Odd number: 5

5. Write an example program to demonstrate inheritance in Dart.

Code:

```
// Base class (Parent)
class Vehicle {
  String brand = "Unknown";

  void start() {
    print("$brand engine started.");
  }

  void stop() {
    print("$brand engine stopped.");
  }
}

// Derived class (Child)
class Car extends Vehicle {
  String model = "Sedan";

  void displayInfo() {
    print("Brand: $brand");
    print("Model: $model");
  }
}

// Another derived class
class Bike extends Vehicle {
  bool hasGear = true;

  void showDetails() {
    print("Brand: $brand");
    print("Has Gear: $hasGear");
  }
}

// Main function
void main() {
  // Object of Car
  Car car = Car();
  car.brand = "Tesla";
  car.displayInfo();
  car.start();
  car.stop();

  print(""); // blank line
}
```

```
// Object of Bike
Bike bike = Bike();
bike.brand = "Yamaha";
bike.showDetails();
bike.start();
bike.stop();
}
```

Output:

Brand: Tesla
Model: Sedan
Tesla engine started.
Tesla engine stopped.

Brand: Yamaha
Has Gear: true
Yamaha engine started.
Yamaha engine stopped.

6. Demonstrate how to implement functions in Dart.

Code:

```
// 1 Simple function (no return type, no parameters)
void greet() {
  print("Hello, welcome to Dart!");
}

// 2 Function with parameters
void displayUser(String name, int age) {
  print("Name: $name, Age: $age");
}

// 3 Function with return type
int add(int a, int b) {
  return a + b;
}

// 4 Function with optional named parameters
void showDetails({String? city, String? college}) {
  print("City: ${city ?? 'Unknown'}");
  print("College: ${college ?? 'Not specified'}");
}

// 5 Arrow function (short form)
int square(int n) => n * n;

// Main function
void main() {
  // calling simple function
  greet();

  // calling function with parameters
  displayUser("Sathwik", 21);

  // calling function with return type
  int sum = add(10, 20);
  print("Sum: $sum");

  // calling function with named parameters
  showDetails(city: "Hyderabad", college: "AI & ML Institute");

  // calling arrow function
  print("Square of 5: ${square(5)}");
}
```

Output:

Hello, welcome to Dart!

Name: Sathwik, Age: 21

Sum: 30

City: Hyderabad

College: AI & ML Institute

Square of 5: 25

7. Write a Dart program to implement the RichText widget.

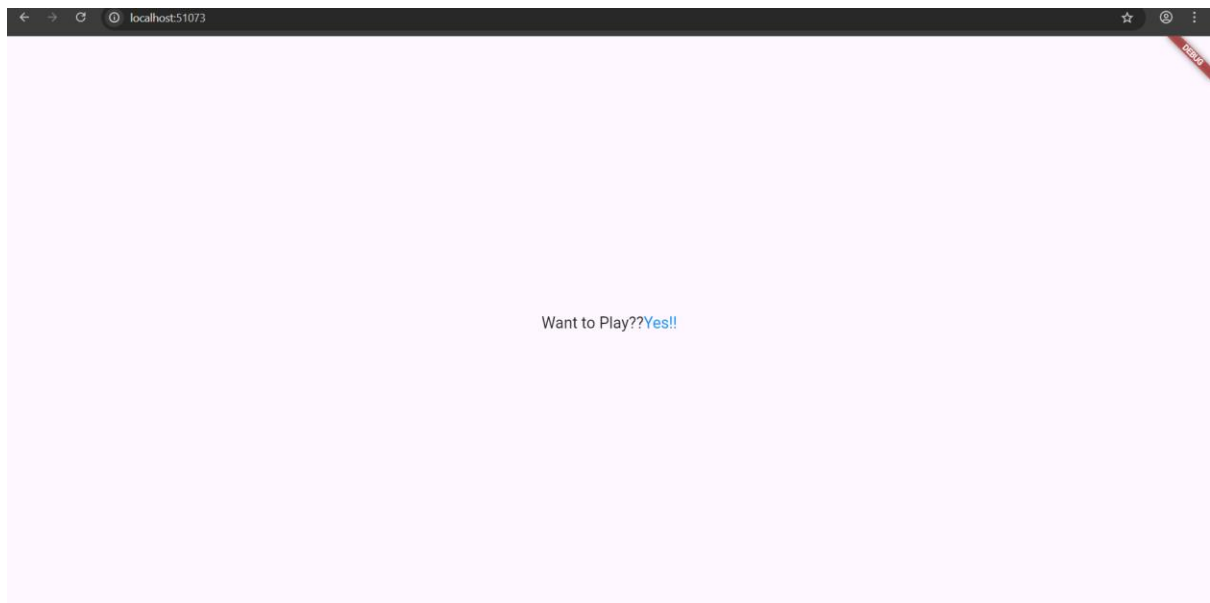
Code:

```
import 'package:flutter/gestures.dart';
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        body: Center(
          child: RichText(
            text: TextSpan(
              text: 'Want to Play??',
              style: TextStyle(fontSize: 20, color: Colors.black87),
              children: <TextSpan>[
                TextSpan(
                  text: 'Yes!!',
                  style: TextStyle(fontSize: 20, color: Colors.blue),
                  recognizer: TapGestureRecognizer()
                    ..onTap={(){}},
                ),
              ],
            ),
          ),
        ),
      ),
    );
  }
}
```

Output:



8. Write a Dart program to display a popup menu button widget.

Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(
          title: Text('Pop Up Button Example'),
          actions: [
            PopupMenuButton(
              onSelected: (value){
                print('Selected $value');
              },
              itemBuilder: (context)=>[
                PopupMenuItem(value:"Option 1",child: Text("Option 1")),
                PopupMenuItem(value:"Option 2",child: Text("Option 2")),
                PopupMenuItem(value:"Option 3",child: Text("Option 3")),
              ]
            ),
          ],
        ),
        body: Center(
          child: Text("Menu on Top Right"),
        ),
      ),
    );
  }
}
```

Output:



9. Write a Dart program to demonstrate a TextField widget.

Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        body: Padding(
          padding: EdgeInsets.all(20.0),
          child: Center(
            child: TextField(
              decoration: InputDecoration(
                border: OutlineInputBorder(),
                labelText: "Enter your name",
                hintText: "Name",
              ),
            ),
          ),
        ),
      ),
    );
  }
}
```

Output:



10. Write a Dart program to insert an image in an application window.

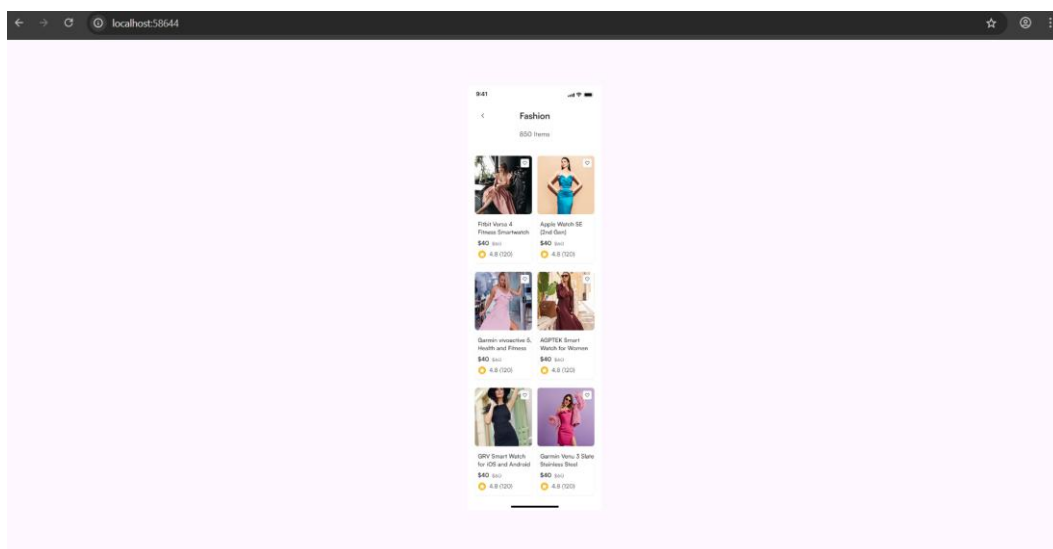
Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        body: Padding(
          padding: EdgeInsets.all(20.0),
          child: Center(
            child: Image.asset(
              'assets/sample.png',
              width: 500,
              height: 600,
            ),
          ),
        ),
      ),
    );
  }
}
```

Output:



11. Write a Dart program to implement the Container widget.

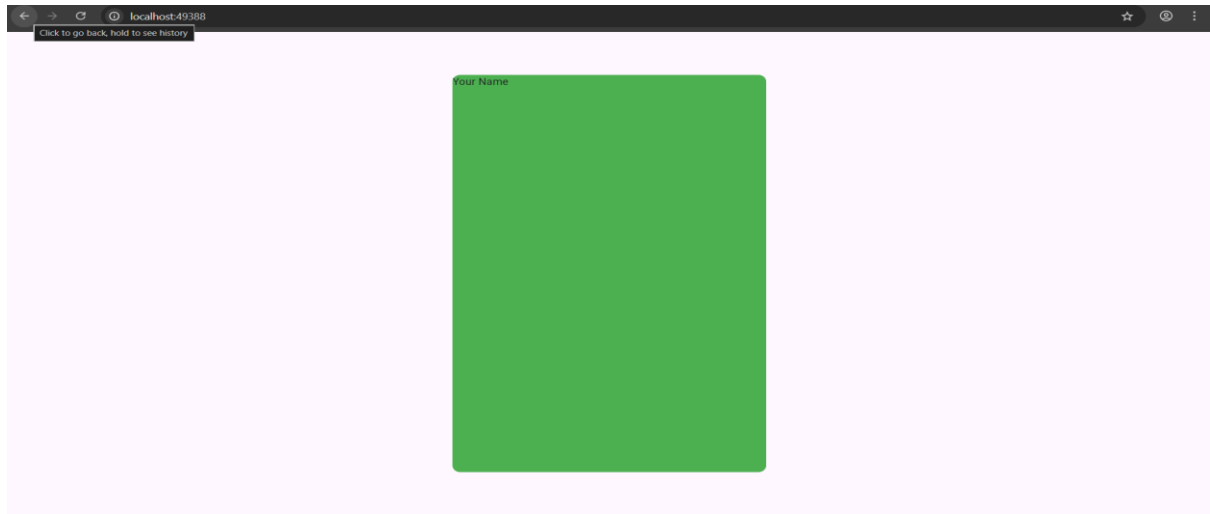
Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        body: Padding(
          padding: EdgeInsets.all(20.0),
          child: Center(
            child: Container(
              width: 400,
              height: 600,
              decoration: BoxDecoration(
                borderRadius: BorderRadius.circular(10),
                color: Colors.green,
              ),
            ),
            child: Text("Your Name"),
          ),
        ),
      ),
    );
  }
}
```

Output:



12. Write a Dart program to demonstrate the Floating Action Button widget.

Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        body: Padding(
          padding: EdgeInsets.all(20.0),
          child: Center(
            child: FloatingActionButton(onPressed: (){}),
            backgroundColor: Colors.red,
            child: Text("data"),
          ),
        ),
      ),
    );
  }
}
```

Output:

